

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1483

Chambers Springs 138 kV
February 24, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1483. The Interconnection Customer proposes a 79.1 MW solar photovoltaic facility interconnecting at the Chambers Springs 138 kV substation in the Pennsylvania area. The request was received on May 28, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 427 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,683,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N DAYTON – JIM BRIDGER 138kV	LAKIN MUNICIPAL 345/138kV TX	105.5	255	5.5
GRAND – SIMMS 138kV Line	LAKIN MUNICIPAL 345/138 kV Autotransformer	104.4	223	5.5
KENNETH FORD – DWP 138kV line	ARCHBALD POWER STATION SUBSTATION – CHAMERS SPRINGS 500kV L	102.6	227	22.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N DAYTON – JIM BRIDGER 138kV	Reconductor line	\$526,000	100% (\$526,000)
GRAND – SIMMS 138kV Line	Reconductor line	\$784,000	100% (\$784,000)
KENNETH FORD – DWP 138kV line	Reconductor line	\$999,000	37% (\$373,000)
	Total allocated to project		\$1,683,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.